Interpreting Recovery in Human–Robot Affinity

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Abstract

A response to a robot can return above its initial level even while an adverse contribution continues to grow. Interpreting such recovery requires distinguishing what a trajectory can exclude from what it can identify. We examine models in which the same positive process contributes to the response and drives an opponent through fixed nonnegative filtering. With zero history, fixed positive observation weights and a log-concave positive driver, the opponent-to-driver ratio cannot decrease, so a return above baseline after a negative excursion is impossible. The argument applies to arbitrary nonnegative delay mixtures and includes a published isolated-dose opponent architecture under its actual decaying input. A saturating two-state construction illustrates one way to escape the restriction while both contributions increase. Fixed nondecreasing scalar observation maps preserve the strict crossing diagnostic, whereas arithmetic averaging need not: an exact example has a recovering mean although neither individual recovers. These conditional comparisons clarify how individual trajectories, exposure history and observation assumptions enter model selection. The examples are synthetic and do not validate a psychological interpretation of the latent states.

Keywords: human–robot interaction; uncanny valley; opponent processes; response recovery; model comparison; log-concavity

1. What a recovered response tells us

When a person's response to a robot improves after an unpleasant interval, the negative response may have weakened. A favorable contribution may also have grown enough to outweigh it. A rating that combines the two cannot, by itself, distinguish these explanations, so an improvement in the rating need not mean that discomfort has subsided.

The question here concerns the development of one person's response during an exposure episode. It differs from Mori's proposed relation between human likeness and affinity [1], and it requires a different account of time from a comparison across separate encounters. Human–robot interaction research already distinguishes eeriness from the unfavorable end of warmth [2]. It has also measured eeriness and likeability separately during repeated interactions: the study by Złotowski and colleagues used an introductory video followed by three interaction phases, while leaving interaction durations unmeasured [3]. Those measurements motivate keeping response dimensions distinct, but they do not establish two additive latent processes or specify how one process drives another.

To compare such dynamical explanations, we consider a response that first rises above its initial level, then falls below it, and later returns above it. We call the final return **recovery**

relative to the initial response. The initial level need not be psychologically neutral, and an upward movement that remains below it is a weaker observation. Figure 1 illustrates the specified pattern using two increasing contributions. The example establishes that a declining opponent is unnecessary; its further value depends on identifying proposed mechanisms that cannot produce the same return.

Suppose, for example, that an analyst represents a favorable process by an exponential approach to a plateau and uses that process to recruit an adverse response through one or more fixed delay stages. Before fitting the number of stages or their rates, the analyst can ask whether the resulting family permits recovery at all. We show that it does not when the model starts from zero internal activity, the output weights remain fixed and the measured response obeys the scalar observation rule specified below. The restriction concerns the proposed family; it does not require the favorable process to have been measured separately. Once a reliable individual return contradicts the family, the explanation must change at the level of its assumptions rather than through a further adjustment of delay parameters.

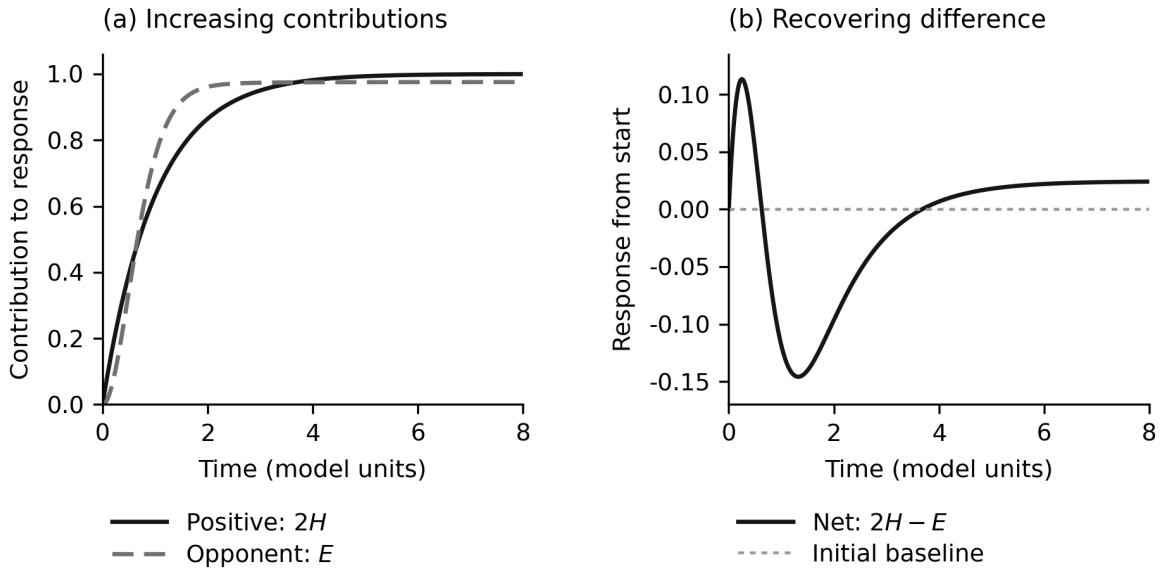


Figure 1: Recovery from increasing contributions. In this synthetic example, the positive contribution 2H and the opponent magnitude E both increase, but their difference first rises above its initial value, becomes negative and eventually returns above it. The equations and analytical sign check are given in Section 3 and Appendix B. Time and response are in model units; the dotted line is the initial response baseline. No participant data are shown.

2. A restriction on fixed delayed opposition

Let $H(t)$ denote a nonnegative process that contributes favorably to a latent response, and let $E(t)$ denote the magnitude of an opposing contribution. Their weighted difference is

$$f(t) = pH(t) - qE(t), \qquad p, q > 0.$$

The weights convert the two contributions to common response units and remain fixed throughout the episode. We take $H(0) = E(0) = 0$, with no activity before onset. Then $f(0) = 0$, so the sign of f specifies whether the latent response is above or below its starting level. A recorded rating $R(t)$ will initially be represented as $R(t) = g(o + f(t))$, where o is a fixed offset and g is

a fixed strictly increasing scalar function. Under that observation assumption, the recorded and latent baseline displacements have the same sign. Section 4 considers its limits.

A simple delayed opponent satisfies $E' = \gamma H - \beta E$, with $\gamma, \beta > 0$. Its value accumulates past input from H, while earlier input has progressively less influence. More generally, a fixed nonnegative filter has the form

$$E(t) = \int_0^\infty k(s) H(t-s)\, ds, \qquad k(s) \geq 0,$$

where $H(u) = 0$ for $u \leq 0$ and the integral is finite. The kernel k fixes the weight given to each delay. For the simple opponent, $k(s) = \gamma e^{-\beta s}$. Fixed nonnegative mixtures and cascades of these filters remain in the same class; replacing the kernel by a nonnegative measure also includes exact delays, as Appendix A explains. There is no extra term carrying activity from an earlier episode.

The shape of the driver determines a useful restriction. Assume $H(t) > 0$ for $t > 0$ and that $\log H$ is concave. Where differentiable, this says that the proportional rate H'/H never increases. It permits both a rise toward a plateau and certain single-peaked responses; it does not require H itself to keep rising. In the proposed plateau model,

$$H(t) = h(1 - e^{-at}), \qquad h, a > 0,$$

the condition follows directly from $H'/H = a/(e^{at} - 1)$. Thus it is a property of the candidate model before any opponent parameters are fitted.

For a fixed delay s, log-concavity means that $H(t-s)$ cannot become smaller *relative to* $H(t)$ as time advances. To see why, the increment $\log H(t) - \log H(t-s)$ is nonincreasing for a concave function, so its negative exponential, $H(t-s)/H(t)$, is nondecreasing. Before the delayed response begins, its ratio is zero. A nonnegative weighted integral of these delayed ratios therefore also cannot decrease:

$$\frac{E(t)}{H(t)} = \int_0^\infty k(s) \frac{H(t-s)}{H(t)}\, ds.$$

The underlying translation-ratio property is established log-concavity mathematics [4]; the argument here uses it directly for a driver, which need not be an integrable probability density. Since

$$\frac{f(t)}{H(t)} = p - q\frac{E(t)}{H(t)},$$

the relative net response is nonincreasing. Once f is nonpositive at a positive time, it cannot later become positive. In particular, the negative-to-positive part of the pattern in Figure 1 already excludes this family. Adding fixed opponent stages cannot remove the restriction while their combined effect remains a nonnegative filter of the same driver.

A published example makes the extent of the comparison concrete. The isolated-dose subsystem of Chou and D'Orsogna has a favorable process driven by an exponentially decaying input, followed by a linearly recruited opponent [5]. Its driver is log-concave, including when the input and response decay rates coincide; Appendix A verifies this using the published equations. The restriction therefore applies to that subsystem without replacing the actual input by a constant one. Its repeated-dose and adaptive extensions change the conditions and are outside

this conclusion. The correspondence concerns the mathematical architecture and supplies no biological identification of human–robot responses.

The exclusion rests on the conjunction of the same-driver relation, log-concavity, zero history, fixed weights and fixed nonnegative filtering, together with the observation assumption. A crossing rejects that conjunction without identifying which part fails. The next example changes the recruitment rule while retaining a zero start and fixed coefficients, allowing us to follow one specific route to recovery.

3. Recovery through a changing balance

In a saturating model, the opponent's current magnitude changes how strongly it responds to further input. Consider the bounded states $H, E \in [0, 1]$ with

$$H' = a - (a + d)H, \qquad E' = bH(1 - E) - cE, \qquad H(0) = E(0) = 0,$$

where $a, d, b, c > 0$. The factor $1 - E$ represents diminishing capacity for further opponent recruitment. Although the coefficients are fixed, the response to the driver is no longer a fixed linear filter: recruitment depends on the opponent's own state. This state architecture has an antecedent in Sontag's bilinear feedforward family [6]. The associated response and its parameter classification are analyzed in the technical note that accompanies the present discussion [7].

The favorable process starts immediately and rises toward $h = a/(a + d)$. Recruitment of the opponent begins only after H becomes positive. Near onset, $H(t) = at + O(t^2)$ and $E(t) = abt^2/2 + O(t^3)$, so every positive pair of output weights gives an initial positive displacement. As the driver grows, the opponent moves toward the instantaneous target $bH/(bH + c)$. That target also increases, and the opponent remains below it, which keeps E increasing throughout the episode. Recovery, if it occurs, must therefore result from a reversal of relative dominance rather than a decline of the opponent.

For Figure 1, choose $a = d = 0.5$, $b = 8$, $c = 0.1$, $p = 2$ and $q = 1$, in arbitrary model units. The driver is $H(t) = (1 - e^{-t})/2$. Its contribution $2H$ begins ahead, but the growing opponent overtakes it before time one. Saturation then slows the opponent's recruitment sufficiently for the favorable contribution to regain dominance. The net response has initial slope one, is negative at $t = 1$, and tends to $1/41 > 0$. Appendix B proves these signs without relying on the numerical plot.

The example shows how recovery can coexist with an increasing opponent, but it does not select saturation as the mechanism behind a measured trace. An additional favorable pathway could also change the balance; a nonzero internal state at onset or coefficients that change during the episode would change the earlier comparison as well. Even the labels attached to H and E require evidence: measuring familiarity and eeriness does not establish that the positively contributing process causes opponent recruitment. The synthetic model demonstrates a possible dynamical explanation whose psychological interpretation remains to be tested.

4. From a latent trajectory to a recorded response

The model comparison uses baseline signs rather than precise response amplitudes. This makes one part of the inference insensitive to the choice of numerical scale. For the fixed strictly increasing map introduced in Section 2,

$$\operatorname{sign}[R(t) - R(0)] = \operatorname{sign} f(t).$$

Consequently a nonlinear rescaling of this kind cannot turn a forbidden baseline return into an allowed one. The statement concerns signs: it does not preserve the ratios used in the proof or establish the shape of the unobserved driver. A more general rating that depends nonlinearly on several states need not satisfy this scalar observation model.

A fixed nondecreasing map, including deterministic quantization, gives a useful one-way conclusion. If the recorded score is strictly above its baseline, the latent score must also be above baseline, and the same implication holds below it. Ties can hide a crossing, but quantization alone cannot manufacture a strict positive–negative–positive pattern. Actual questionnaire responses introduce further questions about noise, changing use of the scale and the comparability of repeated measurements. Those questions need to be addressed before applying a noiseless sign criterion to an individual participant. Appendix C gives a limited finite-sample statement when deterministic error bounds are available; it does not supply a statistical noise model.

Even perfectly known scores can lose the relevant individual structure when they are averaged. Figure 2 gives an exact example on a common numerical scale:

$$f_1(t) = \frac{3}{5}(1 - e^{-t}), \qquad f_2(t) = (1 - e^{-8t}) - \frac{3}{2}(1 - e^{-4t}).$$

The first response remains positive. For the second, putting $x = e^{-4t}$ gives $f_2 = (1-x)(x-1/2)$, so it crosses downward at $t = (\log 2)/4$ and remains negative afterwards. Neither individual therefore exhibits a negative excursion followed by recovery. Yet their equal-weight mean $m = (f_1 + f_2)/2$ satisfies

$$m'(0) = \frac{13}{10} > 0, \qquad m(\log 2) = -\frac{141}{2560} < 0, \qquad \lim_{t\to\infty} m(t) = \frac{1}{20} > 0.$$

The mean has the full recovered excursion. Its late increase reflects the growing contribution of the first trajectory while the second remains negative. The construction shows why a group average cannot substitute for an individual trace when testing an individual-level model. It is an arithmetic example, not a justification for treating arbitrary ordinal ratings as cardinal measurements, and no claim of a newly discovered aggregation effect is needed for this conclusion.

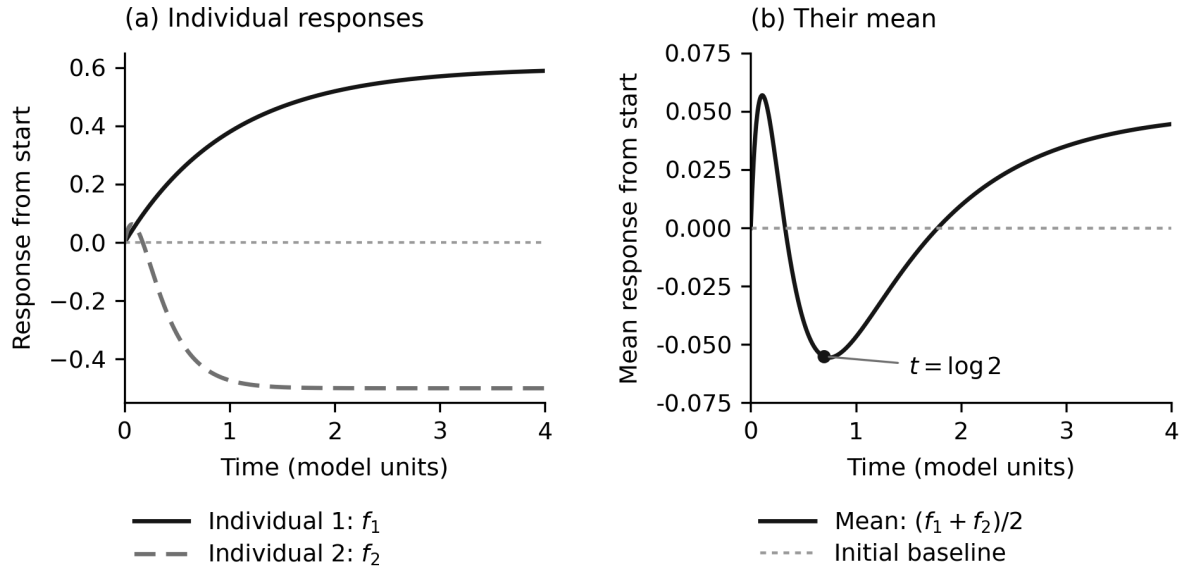


Figure 2: Recovery in a mean without individual recovery. The two synthetic responses in the left panel have no negative excursion followed by a return above their initial baseline. Their equal-weight mean in the right panel does have such a return. The marked value at t = log 2 is exactly $-141/2560$; the positive limiting value is $1/20$. The panels use different vertical scales to show the individual and mean responses clearly. No participant data are shown.

5. Using the comparison

The comparison offers a test of proposed dynamics before a more detailed fit. For the plateau driver and fixed nonnegative opponent filtering, a reliable individual return above baseline rules out the entire family, irrespective of the number of opponent stages or their fitted rates. A successful fit of the saturating model answers a different question: it establishes compatibility with that construction, while leaving competing constructions and the meaning of their latent states unresolved.

Using the test requires keeping the relevant trajectory intact. Individual starting levels and measurement times must be retained, together with information about stimulus changes and previous exposure. A stable robot appearance alone does not establish constant internal coefficients, and a neutral-looking rating does not establish zero latent history. Separate encounters require a rule for carryover between them. These are inputs to the model comparison, so they belong in the proposed account of the experiment rather than being inferred from the shape of the final average.

Evidence	Supported conclusion
Reliable individual return under the stated scalar observation rule	At least one assumption of the fixed-filter family fails.
The same strict signs after a fixed nondecreasing rescaling or noiseless quantization	The crossing diagnostic remains valid; response ratios and effect sizes need not be preserved.
Recovery in an arithmetic mean alone	The aggregate recovers; no individual recovery follows.
A fit of the saturating construction	That model can reproduce the trace; the mechanism and latent labels remain unselected.
No observed return	The observations are compatible with the restriction; neither the mechanism nor absence of crossings between measurements is established.

The useful inference is thus conditional but specific. A recovered trajectory can show that a proposed explanation lacks the dynamics needed to generate it. Determining which alternative explains a person's response requires additional measurements or interventions that distinguish the alternatives. Preserving that separation allows the trajectory to inform model choice without attributing a psychological mechanism to the curve alone.

Appendix A. The filter argument and the published comparator

For a statement that includes exact delays, let H be continuous on $[0,\infty)$, with $H(0) = 0$, $H(t) > 0$ for $t > 0$, and concave $\log H$ on $(0,\infty)$. Extend H by zero to negative time. Let μ be a fixed nonnegative locally finite Borel measure on $[0,\infty)$, and set

$$E(t) = \int_{[0,t]} H(t-s)\,\mu(ds).$$

For fixed $s > 0$, the ratio $H(t-s)/H(t)$ is zero when $0 < t \le s$. For $t > s$, the increment $\log H(t) - \log H(t-s)$ is nonincreasing by concavity, so the ratio is nondecreasing there as well. For $s = 0$ it is identically one at positive times. Integrating this pointwise ordering proves that E/H is nondecreasing. Therefore $f/H = p - qE/H$ is nonincreasing, and $f(t_1) \le 0$ implies $f(t_2) \le 0$ for every $t_2 > t_1 > 0$. Local finiteness and continuity make the integral finite on compact time intervals. A kernel corresponds to $\mu(ds) = k(s)ds$, and an atom places a fixed weight at a specific delay.

In equations (1)–(3) of Chou and D'Orsogna [5], reverse the sign of the negative process and retain the external decay rate before their time rescaling. The isolated zero-history subsystem becomes

$$H' = Ce^{-\delta t} - \alpha H, \qquad E' = \Gamma H - \beta E, \qquad H(0) = E(0) = 0,$$

with positive constants and net response $H - E$. For $\alpha \neq \delta$, put $r = |\alpha - \delta|$ and $v = \min(\alpha, \delta)$. Then

$$H(t) = \frac{C}{r} e^{-vt}(1 - e^{-rt}), \qquad (\log H)'' = -\frac{r^2 e^{rt}}{(e^{rt}-1)^2} < 0.$$

When $\alpha = \delta$, $H(t) = Cte^{-\alpha t}$ and $(\log H)'' = -1/t^2 < 0$. In both cases $E(t) = \Gamma \int_0^t e^{-\beta s} H(t - s)ds$, which has the required nonnegative kernel. This proves the stated restriction for that subsystem and for any fixed nonnegative replacement filter of the same driver, without extending the conclusion to repeated or adaptive inputs.

Appendix B. Analytical checks for the saturating example

The vector field points inward on the boundary of $[0, 1]^2$, apart from permitted tangencies, so the solution stays in that square. Write $T(t) = bH(t)/(bH(t) + c)$. Since H increases, $T'(t) > 0$. The difference $D = T - E$ satisfies $D' = T' - (bH + c)D$ and $D(0) = 0$. Variation of constants gives $D(t) > 0$ for $t > 0$, and hence $E' = (bH + c)D > 0$. Thus both components in Figure 1 increase throughout the episode.

For its selected coefficients, $f'(0) = 1$. To establish a negative intermediate value, let E_0 solve the same opponent equation with $c = 0$. Direct integration gives $E_0(t) = 1-\exp[-4(t-1+e^{-t})]$. The difference $\Delta = E_0 - E$ satisfies $\Delta' = -8H\Delta + E/10$, so $0 \le \Delta(t) \le t/10$. Consequently

$$f(1) \le \frac{1}{10} - e^{-1} + e^{-4/e} < 0.$$

The final inequality can be checked using $e < 11/4$ and the fourth-order Taylor lower bound for $\exp(16/11)$; it does not depend on numerical integration of the state equations. Finally $H \to 1/2$ and $E \to 40/41$, giving $f \to 1/41 > 0$. For the opponent limit, its increasing bounded trajectory has a limit, and the equation forces that limit to be $40/41$ because its derivative would otherwise approach a positive constant. The initial slope, intermediate negative value and positive limit establish the recovered excursion. The broader parameter classification remains in [7].

To identify the antecedent state architecture [6], put $A = a + d$, $X = bH + c$ and $Q = 1 - E$. Then $X' = ab + Ac - AX$ and $Q' = c - XQ$, a constant-input member of Sontag's bilinear feedforward family. The correspondence concerns the states; it does not transfer an adaptation property to the mixed output used here.

Appendix C. A bounded-error sign check

Suppose the recorded samples are $z_i = R(t_i) + e_i$, including baseline index zero, and that deterministic bounds $|e_i| \le \varepsilon_i$ are known. Then $z_i - z_0 > \varepsilon_i + \varepsilon_0$ certifies a positive true baseline displacement, while $z_i - z_0 < -(\varepsilon_i + \varepsilon_0)$ certifies a negative one. Three ordered certified signs positive, negative, positive therefore contradict the fixed-filter family under the observation assumptions. A certified negative-to-positive return already suffices. Statistical confidence intervals would require a separate joint-coverage argument; the deterministic statement provides neither such an argument nor a practical estimator for questionnaire noise.

Relationship to earlier work and research statements

This article is a conceptual and methodological companion to the author's technical preprint [7]. That note gives the two-state model's exact ratio classification and identification calculations; the present article develops a comparison covering arbitrary fixed nonnegative filters, its observation assumptions and an exact averaging example. An earlier explanatory article in The Quantastic Journal introduced the human–robot motivation [8]. The present text distinguishes these model calculations from empirical explanations of the uncanny valley.

All examples are synthetic. No participant experiment or empirical parameter fit is reported. The supporting archive contains editable manuscript sources, the original PNG figures, their

Python reproduction script and numerical checks for the plotted trajectories. The mathematical arguments are given above and in the cited technical note.

OpenAI Astra Ultra assisted with mathematical analysis, literature comparison, drafting and figure code. The author is responsible for the content, interpretations and references. This preprint has not undergone journal peer review.

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